UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	MIDI files store digitized audio samples.	MIDI stores musical instructions, not samples.
A2	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A3	TRUE	A graphics tablet is used to draw directly into a computer.	A tablet is a drawing input device.
A4	TRUE	Keyframes define important positions in an animation.	Keyframes mark major changes.
A5	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A6	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A7	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A8	TRUE	Frame rate is the number of video frames displayed per second.	Frame rate is measured in fps.
A9	TRUE	Tweening creates intermediate frames between keyframes.	Tweening fills the gaps.
A10	TRUE	Cel animation is created by drawing each frame by hand.	Cel animation uses hand-drawn frames.
A11	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A12	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A13	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A14	TRUE	Streaming allows media to begin playing before the whole file is downloaded.	Streaming delivers media continuously.
A15	FALSE	Higher resolution always means a smaller file.	Higher resolution usually increases file size.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	F	Multimedia	integration of text, graphics, audio, video and animation
B2	I	Bitmap	image made of pixels
B3	E	Sampling	converting sound into digital values
B4	K	Storyboard	visual sequence plan
B5	B	MIDI	standard for musical notes and instructions
B6	A	Keyframe	important frame in an animation
B7	N	Scanner	input device that digitises documents
B8	O	Browser	software that displays web pages
B9	G	Amplitude	height of a sound wave
B10	L	HTML	language for web structure
B11	D	Tweening	generating in-between frames
B12	C	Linear multimedia	presentation without user control of sequence
B13	H	Copyright	exclusive rights of a creator
B14	J	Plotter	output device for large drawings
B15	M	Card-based tool	authoring tool organised like pages or cards

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	Display web pages
C2	C	An early model of a product
C3	B	CMYK
C4	B	Space between individual character pairs
C5	D	A simulated environment
C6	A	Plotter
C7	B	Another resource
C8	D	Replace a colour background
C9	B	Links to other content
C10	A	Loudness
C11	B	Scanner
C12	B	A low-fidelity layout sketch
C13	D	Draws alternate lines of a frame
C14	A	Card-based
C15	C	Assemble multimedia elements
C16	C	Text, graphics, audio, video and animation
C17	D	Temporary working memory
C18	C	A visual sequence plan
C19	A	Printed output
C20	D	Analog sound to digital values
C21	D	Pixels
C22	A	Small strokes at character ends
C23	B	Merging of media and computing
C24	D	Multimedia with hyperlinks
C25	D	A video compression family
C26	B	Number of pixels in an image
C27	C	A cinema presentation
C28	C	One shape into another
C29	C	Along a predefined route
C30	A	Screen display

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PART D - WORKED ANSWERS

QUESTION 1

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

QUESTION 2

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(5 marks)

QUESTION 3

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

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QUESTION 4

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

END OF MARKING SCHEME

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